CS 70 Discrete Mathematics and Probability Theory
Fall 2025 Course Notes Note 0

Review of Sets and Mathematical Notation

A **set** is a well defined collection of objects. These objects are called **elements** or **members** of the set, and they can be anything, including numbers, letters, people, cities, and even other sets. By convention, sets are usually denoted by capital letters and can be described or defined by listing its elements and surrounding the list by curly braces. For example, we can describe the set A to be the set whose members are the first five prime numbers, or we can explicitly write: $A = \{2,3,5,7,11\}$. If x is an element of A, then we write $x \in A$. Similarly, if y is not an element of A, then we write $y \notin A$. Two sets A and B are said to be **equal**, written as $A = B$, if they have the same elements. The order and repetition of elements do not matter, so {red, white, blue} = {blue, white, red} = {red, white, white, blue}. Sometimes, more complicated sets can be defined by using a different notation. For example, the set of all rational numbers, denoted by $\mathbb{Q}$, can be written as: $\{\frac{a}{b} \mid a,b \text{ are integers}, b \neq 0\}$. In English, this is read as "the set of all fractions such that the numerator is an integer and the denominator is a non-zero integer."

Cardinality

We can also talk about the size of a set, or its **cardinality**. If $A = \{1,2,3,4\}$, then the cardinality of A, denoted by $|A|$, is 4. It is possible for the cardinality of a set to be 0. There is a unique such set, called the **empty set**, denoted by the symbol $\emptyset$. A set can also have an infinite number of elements, such as the set of all integers, prime numbers, or odd numbers.

Subsets and Proper Subsets

If every element of a set A is also in set B, then we say that A is a **subset** of B, written $A \subseteq B$. Equivalently we can write $B \supseteq A$, or B is a superset of A. A **proper subset** is a set A that is strictly contained in B, written as $A \subset B$, meaning that A excludes at least one element of B. For example, consider the set $B = \{1,2,3,4,5\}$. Then $\{1,2,3\}$ is both a subset and a proper subset of B, while $\{1,2,3,4,5\}$ is a subset but not a proper subset of B. Here are a few basic properties regarding subsets:

- The empty set, denote by $\{\}$ or $\emptyset$, is a proper subset of any nonempty set A: $\{\} \subset A$.
- The empty set is a subset of every set B: $\{\} \subseteq B$.
- Every set A is a subset of itself: $A \subseteq A$.

Intersections and Unions

The **intersection** of a set A with a set B, written as $A \cap B$, is the set containing all elements which are in both A and B. Two sets are said to be **disjoint** if $A \cap B = \emptyset$. The **union** of a set A with a set B, written as $A \cup B$, is the set of all elements which are in either A or B or both. For example, if A is the set of all positive even numbers, and B is the set of all positive odd numbers, then $A \cap B = \emptyset$, and $A \cup B = \mathbb{Z}^+$, or the set of all positive integers. Here are a few properties of intersections and unions:

- $A \cup B = B \cup A$
- $A \cup \emptyset = A$

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注 0

集合与数学符号回顾

集合是定义明确的对象集合。这些对象被称为集合的元素或成员，它们可以是任何事物，包括数字、字母、人、城市，甚至其他集合。按照惯例，集合通常用大写字母表示，可以通过列出其元素并用花括号括起来来描述或定义。例如，我们可以将集合 A 描述为由前五个质数组成的集合，或者我们可以明确地写成：A = {2, 3, 5, 7, 11}。如果 x 是 A 的元素，则记作 x ∈ A。类似地，如果 y 不是 A 的元素，则记作 y ≠ A。如果两个集合 A 和 B 具有相同的元素，则称它们相等，记作 A = B。元素的顺序和重复并不重要，因此 {红, 白, 蓝} = {蓝, 白, 红} = {红, 白, 白, 蓝}。有时，更复杂的集合可以使用不同的符号来定义。例如，所有有理数的集合，记为 Q，可以写成：{ | a, b 为整数，b ≠ 0}。用英语来说，这读作“所有分子为整数且分母为非零整数的分数的集合”。

基数

我们还可以讨论集合的大小，或者说它的基数。如果 A = {1, 2, 3, 4}，那么 A 的基数，记为 |A|，为 4。集合的基数可以为 0。这样的集合是唯一的，称为空集，记为 /0。集合也可以包含无限多个元素，例如所有整数、素数或奇数构成的集合。

子集和真子集

如果集合 A 中的每个元素都属于集合 B，则称 A 是 B 的子集，记作 A ⊆ B。等价地，我们可以记作 B ⊇ A，或者说 B 是 A 的超集。真子集是指严格包含于集合 B 的集合 A，记作 A ⊂ B，这意味着 A 至少排除 B 中的一个元素。例如，考虑集合 B = {1, 2, 3, 4, 5}。那么 {1, 2, 3} 既是 B 的子集又是 B 的真子集，而 {1, 2, 3, 4, 5} 是 B 的子集，但不是 B 的真子集。以下是关于子集的一些基本性质：

- 空集，记为 {} 或 /0，是任何非空集合 A 的真子集：{} ⊂ A。
- 空集是每个集合 B 的子集：{} ⊆ B。
- 每个集合 A 都是它自身的子集：A ⊆ A。

交集和并集

集合 A 与集合 B 的交集，记作 A ∩ B，是包含所有同时属于 A 和 B 的元素的集合。如果 A ∩ B = ∅，则称两个集合不相交。集合 A 与集合 B 的并集，记作 A ∪ B，是包含所有属于 A 或 B 或同时属于两者的元素的集合。例如，如果 A 是所有正偶数的集合，B 是所有正奇数的集合，则 A ∩ B = ∅，且 A ∪ B = Z，即所有正整数的集合。以下是交集和并集的一些性质：

- A ∪ B = B ∪ A
- A ∪ /0 = A

- $A \cap B = B \cap A$
- $A \cap \emptyset = \emptyset$

Complements

If A and B are two sets, then the **relative complement** of A in B, or the **set difference** between B and A, written as $B - A$ or $B \setminus A$, is the set of elements in B, but not in A: $B \setminus A = \{x \in B \mid x \notin A\}$. For example, if $B = \{1,2,3\}$ and $A = \{3,4,5\}$, then $B \setminus A = \{1,2\}$. For another example, if $\mathbb{R}$ is the set of real numbers and $\mathbb{Q}$ is the set of rational numbers, then $\mathbb{R} \setminus \mathbb{Q}$ is the set of irrational numbers. Here are some important properties of complements:

- $A \setminus A = \emptyset$
- $A \setminus \emptyset = A$
- $\emptyset \setminus A = \emptyset$

Significant Sets

In mathematics, some sets are referred to so commonly that they are denoted by special symbols. These include:

- $\mathbb{N}$ denotes the set of all natural numbers: $\{0,1,2,3,...\}$.
- $\mathbb{Z}$ denotes the set of all integer numbers: $\{\ldots,-2,-1,0,1,2,\ldots\}$.
- $\mathbb{Q}$ denotes the set of all rational numbers: $\{\frac{a}{b} \mid a,b \in \mathbb{Z},\ b \neq 0\}$.
- $\mathbb{R}$ denotes the set of all real numbers.
- $\mathbb{C}$ denotes the set of all complex numbers.

Products and Power Sets

The **Cartesian product** (also called the **cross product**) of two sets A and B, written as $A \times B$, is the set of all pairs whose first component is an element of A and whose second component is an element of B. In set notation, $A \times B = \{(a,b) \mid a \in A, b \in B\}$. For example, if $A = \{1,2,3\}$ and $B = \{u,v\}$, then $A \times B = \{(1,u),(1,v),(2,u),(2,v),(3,u),(3,v)\}$. And $\mathbb{N} \times \mathbb{N} = \{(0,0),(1,0),(0,1),(1,1),(2,0),\ldots\}$ is the set of all pairs of natural numbers. Given a set S, the **power set** of S, denoted by $\mathscr{P}(S)$, is the set of all subsets of S: $\{T \mid T \subseteq S\}$. For example, if $S = \{1,2,3\}$, then the power set of S is: $\mathscr{P}(S) = \{\{\},\{1\},\{2\},\{3\},\{1,2\},\{1,3\},\{2,3\},\{1,2,3\}\}$. Note that, if $|S| = k$, then $|\mathscr{P}(S)| = 2^k$. [Why?]

Mathematical Notation

Sums and Products

There is a compact notation for writing sums or products of large numbers of items. For example, to write $1+2+\cdots+n$, without having to say "dot dot dot", we write $\sum_{i=1}^{n} i$. More generally we can write the sum $f(m)+f(m+1)+\cdots+f(n)$ as $\sum_{i=m}^{n} f(i)$. Thus, for example, $\sum_{i=5}^{n} i^2 = 5^2+6^2+\cdots+n^2$.

Analogously, to write the product $f(m)f(m+1)\cdots f(n)$ we use the notation $\prod_{i=m}^{n} f(i)$. For example, $\prod_{i=1}^{n} i = 1 \cdot 2 \cdots n$ is the product of the first n positive integers.

Universal and Existential Quantifiers

- A ∩ B = B ∩ A
- A ∩ /0 = /0

互补

如果 A 和 B 是两个集合，那么 A 在 B 中的相对补集，或者说 B 与 A 的集合差集，记作 B − A 或 B \ A，是属于 B 但不属于 A 的元素的集合：B \ A = {x ∈ B | x ≠ A}。例如，如果 B = {1, 2, 3} 且 A = {3, 4, 5}，那么 B \ A = {1, 2}。再举一个例子，如果 R 是实数集，Q 是有理数集，那么 R \ Q 是无理数集。以下是补集的一些重要性质：

- A \ A = /0
- A \ /0 = A
- ∅ \ A = /0

重要集合

在数学中，有些集合的使用频率非常高，以至于它们用特殊的符号来表示。这些集合包括：

- N 表示所有自然数的集合：{0, 1, 2, 3, ...}。
- Z 表示所有整数的集合：{..., -2, -1, 0, 1, 2, ...}。
- Q 表示所有有理数的集合：{ | a, b ∈ Z, b′= 0}。
- R 表示所有实数的集合。
- C 表示所有复数的集合。

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两个集合 A 和 B 的笛卡尔积（也称叉积），记作 A × B，是所有满足以下条件的元素对的集合：第一个分量是 A 的元素，第二个分量是 B 的元素。用集合符号表示，A × B = {(a, b) | a ∈ A, b ∈ B}。例如，如果 A = {1, 2, 3} 且 B = {u, v}，则 A × B = {(1, u), (1, v), (2, u), (2, v), (3, u), (3, v)}。N × N = {(0, 0), (1, 0), (0, 1), (1, 1), (2, 0), ...} 是所有自然数对的集合。给定一个集合 S，S 的幂集，记作 P(S)，是 S 的所有子集的集合：{T | T ⊆ S}。例如，如果 S = {1, 2, 3}，则 S 的幂集为：P(S) = {{}, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}, {1, 2, 3}}。注意，如果 |S| = k，则 |P(S)| = 2。[为什么？]

数学符号

和与积

对于大量元素的和或积，有一种简洁的表示方法。例如，要表示 1 + 2 + ・・・ + n，而不用写“点点点”，我们可以写成 Σ i。更一般地，我们可以将和 f(m) + f(m + 1) + ・・・ + f(n) 写成 Σ f(i)。因此，例如，Σ i = 5 + 6 + ・・・ + n。

类似地，为了表示乘积 f(m) f(m+1) ・・・ f(n)，我们使用符号 ∏ f(i)。例如，∏ i = 1 ・ 2 ・・・ n 表示前 n 个正整数的乘积。

全称量词和存在量词

Consider the statement: For all natural numbers n, n^2+n+41 is prime. Here, n is quantified to any element of the set $\mathbb{N}$ of natural numbers. In notation, we write $(\forall n \in \mathbb{N})(n^2+n+41 \text{ is prime})$. Here we have used the *universal quantifier* $\forall$ ("for all"). Is the statement true? If you try to substitute small values of n, you will notice that n^2+n+41 is indeed prime for those values. But if you think harder, you can find larger values of n for which it is not prime. Can you find one? So the statement $(\forall n \in \mathbb{N})(n^2+n+41 \text{ is prime})$ is false.

The *existential quantifier* $\exists$ ("there exists") is used in the following statement: $(\exists x \in \mathbb{Z})(x < 2 \text{ and } x^2 = 4)$. The statement says there is an integer x which is less than 2, but its square is equal to 4. This is a true statement (take $x = -2$).

We can also write statements using both kinds of quantifiers:

1. $(\forall x \in \mathbb{Z})(\exists y \in \mathbb{Z})(y > x)$
2. $(\exists y \in \mathbb{Z})(\forall x \in \mathbb{Z})(y > x)$

The first statement says that, given an integer, we can find a larger one. The second statement says something very different: that there is a largest integer! The first statement is true, the second is not.

考虑以下命题：对于所有自然数 n，n + n + 41 都是素数。这里，n 被量化为自然数集 N 中的任意元素。用符号表示为 (∀n ∈ N)(n + n + 41 是素数)。这里我们使用了全称量词 ∀（“对于所有”）。这个命题正确吗？如果你尝试代入较小的 n 值，你会发现对于这些值，n + n + 41 确实是素数。但是，如果你更深入地思考，你会发现可以找到更大的 n 值，使得 n + n + 41 不是素数。你能找到这样的值吗？因此，命题 (∀n ∈ N)(n + n + 41 是素数) 是错误的。

存在量词 ∃（“存在”）用于以下语句：(∃x ∈ Z)(x < 2 且 x = 4)。该语句表示存在一个小于 2 的整数 x，且 x 的平方等于 4。这是一个真命题（取 x = -2）。

我们也可以使用这两种量词来编写语句：

1. (∀x ∈ Z)(∃y ∈ Z)(y > x)
2. (∃y ∈ Z)(∀x ∈ Z)(y > x)

第一句话说，给定一个整数，我们可以找到一个更大的整数。第二句话则完全不同：它说存在一个最大的整数！第一句话是正确的，第二句话是错误的。